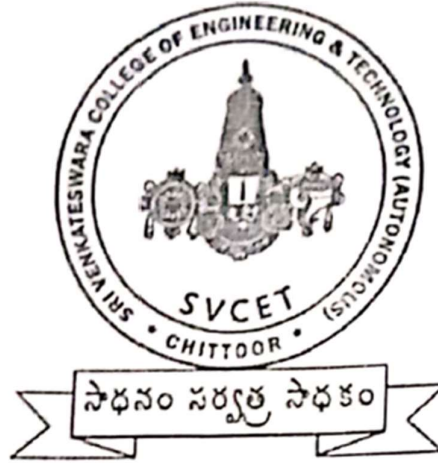


SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(Autonomous)



NAME OF THE STUDENT :

ROLL NUMBER :

BRANCH & YEAR : CSD-C or CSM-C (ASS-3)

SUBJECT WITH CODE : BASIC ELECTRICAL & ELECTRONICS

TOPIC : Energy resources ,Electricity bill and
safety measures

SUBMITTED ON :

SUBMITTED TO :

1) Compare conventional and non conventional energy resources ?

Conventional	Non Conventional
Conventional sources of energy are those sources which have been use since the early times.	Non-conventional sources of energy have generally been identified in the recent past.
They are exhaustible except hydro-energy.	They are inexhaustible.
They cause pollution when used as they emit smoke and ash.	Generally these are pollution-free.
Their generation and use involve huge expenditure.	Low expenditure required.
Very expensive to maintain, store, transmit as they are carried over long distances through transmission grids.	Less expensive due to local use and easy maintenance.

Examples are – coal, natural gas, water, fire-wood.

Examples are – geothermal energy, solar energy, wind energy, tidal energy, biogas energy, nuclear energy.

2) Explain tariff ,two part tariff ,calculation of electricity bill for domestic consumers .

TARIFF

The rate at which electrical energy is supplied to a consumer is known as tariff. Although tariff should include the total cost of producing and supplying electrical energy plus the profit, yet it cannot be the same for all types of consumers. It is because the cost of producing electrical energy depends to a considerable extent upon the magnitude of electrical energy consumed by the user and his load conditions. Therefore, in all fairness, due consideration has to be given to different types of consumers (e.g., industrial, domestic and commercial) while fixing the tariff. This makes the problem of suitable rate making highly complicated.

Objectives of tariff:

- Recovery of cost of producing electrical energy at the power station.
- Recovery of cost on the capital investment in transmission and distribution systems.
- Recovery of cost of operation and maintenance of supply of electrical energy e.g., metering equipment, billing etc.
- A suitable profit on the capital investment

Two-part tariff:

When the rate of electrical energy is charged on the basis of maximum demand of the consumer and the units consumed, it is called a two-part tariff.

In two-part tariff, the total charge to be made from the consumer is split into two components viz., fixed charges and running charges. The fixed charges depend upon the maximum demand of the consumer while the running charges depend upon the number of units consumed by the consumer. Thus, the consumer is charged at a certain amount per kW of maximum demand plus a certain amount per kWh of energy consumed i.e.

$$\text{Total charges} = \text{Rs } (b \times \text{kW} + c \times \text{kWh})$$

where, b = charge per kW of maximum demand

c = charge per kWh of energy consumed

This type of tariff is mostly applicable to industrial consumers who have appreciable maximum demand.

Advantages:

- It is easily understood by the consumers.
- It recovers the fixed charges which depend upon the maximum demand of the consumer but are independent of the units consumed.

Disadvantages:

- The consumer has to pay the fixed charges irrespective of the fact whether he has consumed or not consumed the electrical energy.
- There is always error in assessing the maximum demand of the consumer.

CALCULATION OF ELECTRICITY BILL FOR DOMESTIC CONSUMERS

Calculation of electricity bill for low tension domestic consumer is as follows.

The electricity bill consists of two components: fixed charges and variable charges (running charges). It can be expressed as:

$$\text{Total Electricity Bill} = [A \text{ (kW)} + B \text{ (kWh)}] + \text{Tax}$$

Where, Fixed charges - A = charge per kW of max demand

$$A = \text{Total kW} \times \text{charge per kW}$$

Example: if the sanctioned load is 3KW then $A = [1 \times 85 + 2 \times 95] = 275 \text{rs}$

(Note: For 1kw it is 85 rs and above 1kw it 95 rs per kw)

Where Variable charges - B = charge per kwh of energy consumed.

$$B = \text{No of units consumed} \times \text{rate per unit}$$

Example: If the no of units consumed is 120 units then

$$B = [50 \times 4.1 + 50 \times 5.55 + 20 \times 7.1] = 624 \text{ rs}$$

(Note: For 0- 50 units – 4.1 rs per unit, 50- 100 units – 5.55 rs , 100- 200 units – 7.1rs)

Therefore Total Electricity bill for given example is

$$= 275 + 624 + \text{Tax.}$$

3) Explain personal safety measures and types of earthing?

PERSONAL SAFETY MEASURES

Electric shock and precautions

An electric shock is the sudden discharge of electricity through a part of the body when a person comes in contact with electrical equipment.

The factors affecting the severity of shock are

1. Magnitude of the current through the body
2. Path of the current through the body
3. Time for which current is passed through the body
4. Frequency of the current
5. Physical and physiological condition of the person.

Precautions against Electric shock

- Avoid water at all times when working with electricity. Never touch or try repairing any electrical equipment or circuits with wet hands. It increases the conductivity of the electric current.
- Never use equipment with damaged insulation. The insulation of conductors must be proper and in good condition.
- Earth connection should be maintained in proper condition
- Use of the fuses and cables of proper rating.
- Use the rubber soled shoes while working.
- Megger tests should be done to check the insulation.
- Never touch two different terminals at the same time.
- Never remove the plug by pulling wire.
- The sockets should be placed at a proper height
- Switch off supply and remove the fuses before starting the work with any installation.
- Always use insulated screw drivers, and line testers.

Earthing :

Connection of the body of electric equipment to the general mass of the earth by wire of negligible resistance is called **Earthing**. It brings the body of the equipment to the zero potential during electric shock.

Necessity of Earthing

1. To protect the human beings from danger of shock in case they come in contact with the charged frame due to defective insulation.
2. It guarantees the safety of electrical appliances and devices from the excessive amount of electric current.
3. It protects the appliances from high voltage surges and lightning discharge.
4. It provides an alternative path for leakage of current hence protects the equipment.
5. It keeps the voltage constant in the healthy phase
6. It protects the Electric system and buildings from lightning.
7. It avoids the risk of fire in the electrical installation system.
8. To maintain the line voltage constant under unbalanced load condition.

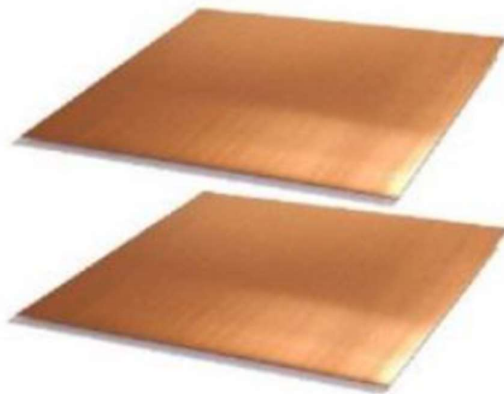
TYPES OF EARTHING

There are different types of earthing system mostly used. They are

- i) Plate earthing ii) Pipe earthing iii) Rod earthing

PLATE EARTHING:

The plate material is of either copper or galvanized iron is used in the plate earthing system. This plate selected should of certain specified dimensions, which is placed inside the earth at a depth less than 3 meters from the bottom. This plate is connected to the electrical conductors to divert the electric charge inside the earth. The diagram of Plate Earthing is given below.



Earthing plate

PIPE EARTHING:

A galvanized iron pipe is selected should be such that it has holes pierced at regular intervals and the pipe is narrow at the bottom end.

A clamp is attached to the G.I pipe to which an earth wire is connected. This pipe diverts the electrical conductance inside the earth. Pipe Earthing is shown in the figure.



earthing pipe

The pipe is placed in the earth pit at a depth not less than 3 meters. The space inside the G.I pipe is filled by the alternate layers of salt and charcoal up to the clamp level.

The top portion of the G.I pipe is left open for maintenance when required by constructing a chamber with bricks. Water is poured into the G.I pipe to maintain earthing resistance within the specified limits.

A galvanized steel and a pipe that has holes at regular intervals are kept inside the earth. Keeping in view its low-cost Pipe earthing is commonly used for all domestic purposes.

ROD EARTHING:

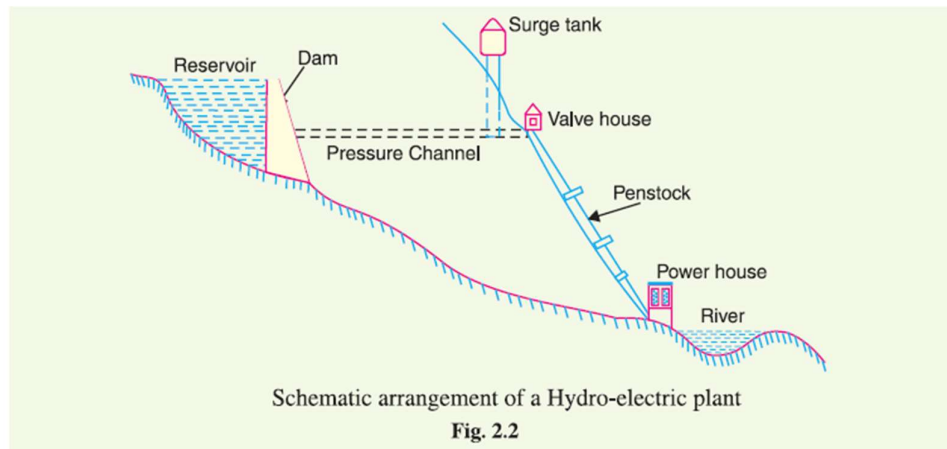
This type of earthing is similar to pipe earthing, but a rod made of galvanized steel is used in this case. The rod used for this purpose is buried inside the earth at a certain depth. As it is of low resistive material, the short circuit current will be diverted to the ground safely. The rod earthing diagram is given the following figure.



Earthing rod

4) Explain hydel and nuclear power plant with a layout?

Ans:



2.10 Constituents of Hydro-electric Plant

The constituents of a hydro-electric plant are (1) hydraulic structures (2) water turbines and (3) electrical equipment. We shall discuss these items in turn.

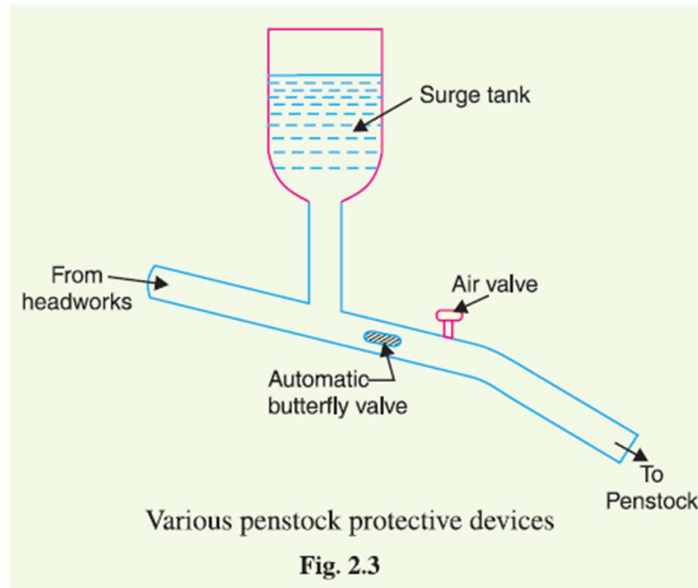
1. Hydraulic structures. Hydraulic structures in a hydro-electric power station include dam, spillways, headworks, surge tank, penstock and accessory works.

(i) **Dam.** A dam is a barrier which stores water and creates water head. Dams are built of concrete or stone masonry, earth or rock fill. The type and arrangement depends upon the

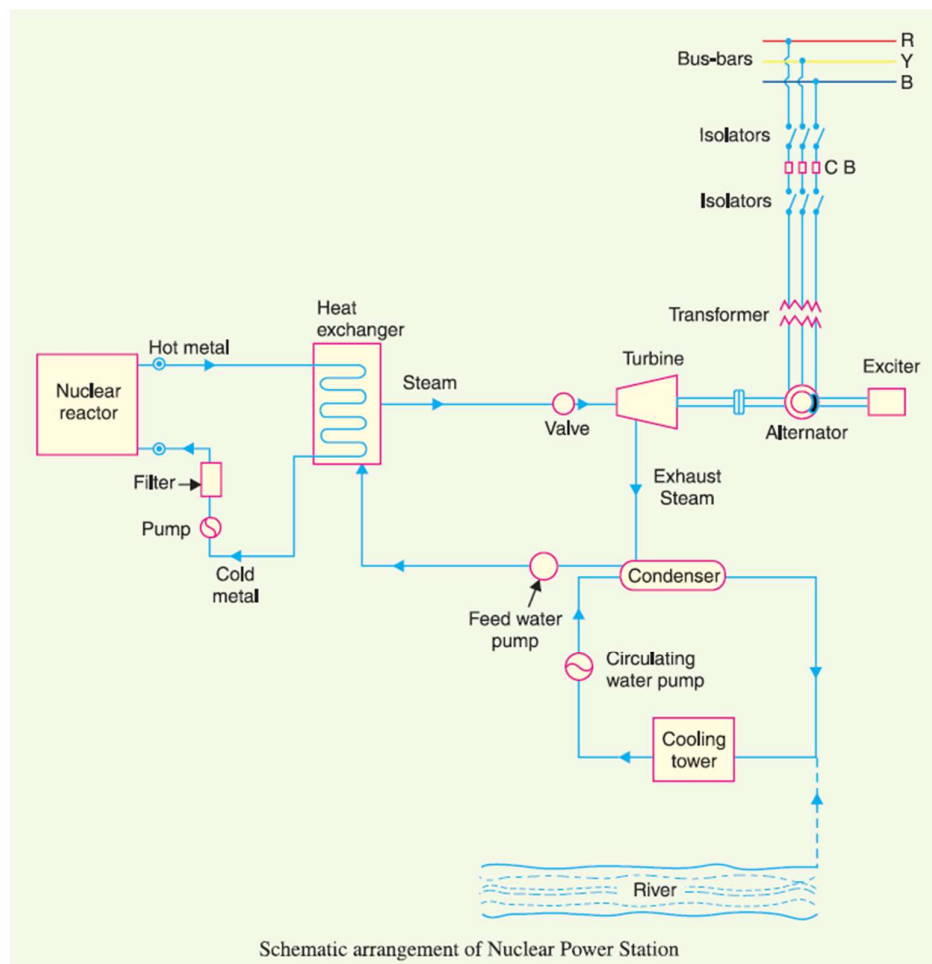
topography of the site. A masonry dam may be built in a narrow canyon. An earth dam may be best suited for a wide valley. The type of dam also depends upon the foundation conditions, local materials and transportation available, occurrence of earthquakes and other hazards. At most of sites, more than one type of dam may be suitable and the one which is most economical is chosen.

(ii) **Spillways.** There are times when the river flow exceeds the storage capacity of the reservoir. Such a situation arises during heavy rainfall in the catchment area. In order to discharge the surplus water from the storage reservoir into the river on the down-stream side of the dam, spillways are used. Spillways are constructed of concrete piers on the top of the dam. Gates are provided between these piers and surplus water is discharged over the crest of the dam by opening these gates.

(iii) **Headworks.** The headworks consists of the diversion structures at the head of an intake. They generally include booms and racks for diverting floating debris, sluices for by-passing debris and sediments and valves for controlling the flow of water to the turbine. The flow of water into and through headworks should be as smooth as possible to avoid head loss and cavitation. For this purpose, it is necessary to avoid sharp corners and abrupt contractions

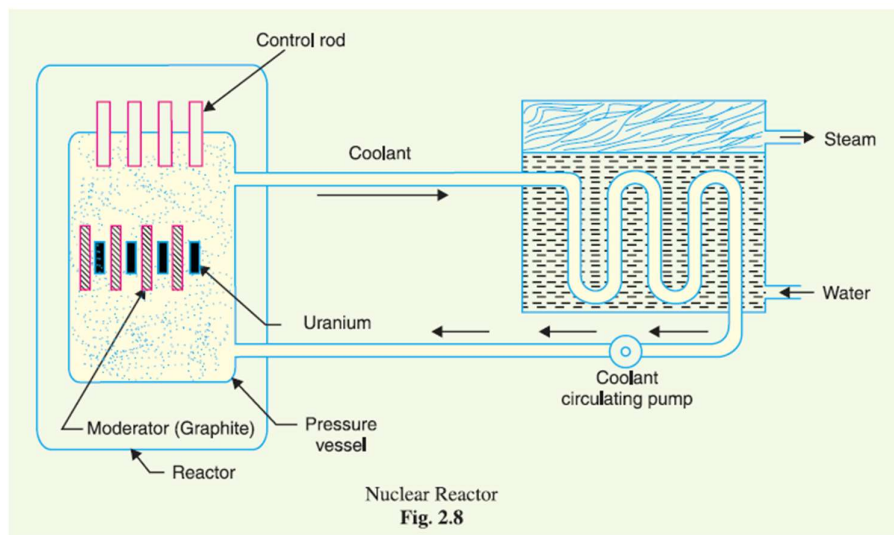


Nuclear power station:



- (i) **Nuclear reactor.** It is an apparatus in which nuclear fuel (U^{235}) is subjected to nuclear fission. It controls the *chain reaction** that starts once the fission is done. If the chain reaction is not controlled, the result will be an explosion due to the fast increase in the energy released.

A nuclear reactor is a cylindrical stout pressure vessel and houses fuel rods of Uranium, moderator and control rods (See Fig. 2.8). The fuel rods constitute the fission material and release huge amount of energy when bombarded with slow moving neutrons. The moderator consists of graphite rods which enclose the fuel rods. The moderator slows down the neutrons before they bombard the fuel rods. The control rods are of cadmium and are inserted into the reactor. Cadmium is strong neutron absorber and thus regulates the supply of neutrons for fission. When the control rods are pushed in deep enough, they absorb most of fission neutrons and hence few are available for chain reaction which, therefore, stops. However, as they are being withdrawn, more and more of these fission neutrons cause fission and hence the *intensity* of chain reaction (or heat produced) is increased. Therefore, by pulling out the control rods, power of the nuclear reactor is increased, whereas by pushing them in, it is reduced. In actual practice, the lowering or raising of control rods is accomplished automatically according to the requirement of load. The heat produced in the reactor is removed by the coolant, generally a sodium metal. The coolant carries the heat to the heat exchanger.



- (ii) **Heat exchanger.** The coolant gives up heat to the heat exchanger which is utilised in raising the steam. After giving up heat, the coolant is again fed to the reactor.

5) Explain solar and wind power with a neat sketch?

Ans :

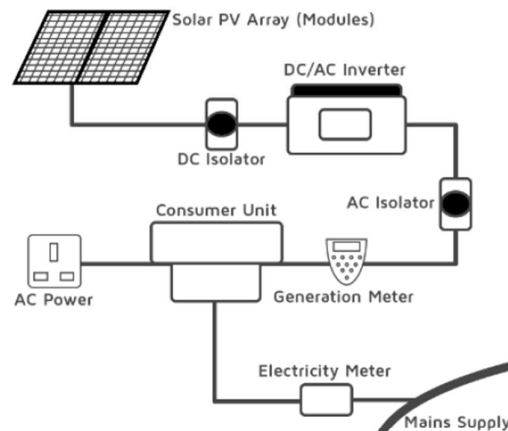
LAYOUT AND OPERATION OF SOLAR POWER GENERATION SYSTEM

Solar photovoltaic (PV) panels use cells containing a semi-conductor material to capture the sun's energy and convert solar radiation into electricity. The most commonly used semi-conductor material is silicon, which is an abundant natural resource found in sand. When light strikes the cell a certain amount of energy is absorbed within the semiconductor material, knocking electrons, the negatively charged particles that form the basis of electricity, loose.

Most PV cells have two layers of semi-conductor material, one positively charged and one negatively charged. When light shines on the semi-conductor the electric field across the junction between these two layers causes electricity to flow, generating direct current (DC). By placing metal contacts on the top and bottom of the PV cell, we can draw that current off for external use.

Solar PV electric panels do not require bright sunlight in order to operate, meaning that you can still generate electricity on cloudy days, however in general the greater the intensity of light the higher the flow of electricity. Although, due to the reflection of sunlight, days with slight cloud can result in higher energy yields than days with a completely cloudless sky.

It is important to realise that you can only use your free solar electricity when it is being generated – so unless you also invest in batteries to store power for use in the evenings and at night, you will need to pay for your energy use as normal when the panels are not producing electric.



A typical grid connected roof mounted system consists of a number of components, as shown in the diagram.

Direct Current (DC) needs to be converted into alternating current (AC) so it can be used in a domestic building; this is performed by an inverter. The AC electricity then passes via the generation meter, which measures how much electricity has been created, and on to the consumer unit where it can be fed into the property for use or exported back to the national grid via the electricity meter.

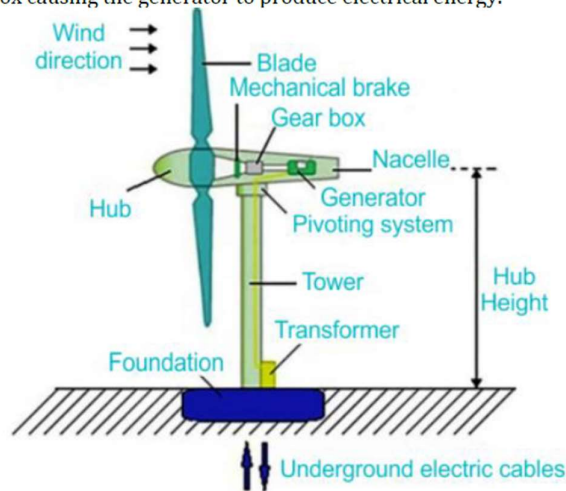
LAYOUT AND OPERATION OF WIND POWER GENERATION SYSTEM

Wind Power plants are a collection of wind turbines either horizontal or vertical type. These turbines collect the energy individually and are connected to a common plant. The wind turbine is also similar to the normal turbine, as it converts kinetic energy into mechanical energy. And they are designed in such a way that the height and length of the blades are maintained at some ratio.

The wind turbine works on the principle of conversion of kinetic energy of wind to mechanical energy used to rotate the blades of a fan connected to an electric generator. When the wind or air touches the blades (or) vanes of the windmill it the air pressure can be uneven, higher on one side of the blade and lower on the other. Hence, uneven pressure causes the blades to spin around the center of the turbine. The turbine does not operate at wind speeds above 55 mph with the use of the controller.

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The rotor shaft of the turbine (ie., low speed and high speed) is interlinked with the gearbox which converts the speed from 30 to 60 rpm into 1000 to 1800 rpm. As the gearbox consists of gears, to transmit mechanical energy. These speeds are most suitable to the generator for the generation of electricity. When the rotor of the turbine rotates it drives a generator through a setup gearbox causing the generator to produce electrical energy.



- **Blades** are usually made of fiberglass or balsa wood. Most turbines have either two or three blades.
- **Rotor:** It includes the blades and the hub together. The blades spin the rotor, which is attached to a shaft that transfers the torque it creates into the gearbox. The rotor provides pitch regulation for power output optimization and control. Its speed is variable to maximize aerodynamic efficiency.

- **Pitch** turns blades out of the wind to control the rotor speed and keep the rotor from turning in winds that are too high or too low to produce electricity.
- **Brake** is a disc that can be applied aerodynamically, electrically, or hydraulically to stop the rotor in emergencies. A brake shuts down the turbine if the winds become strong enough to impact the turbine's internal components.
- **Low-Speed Shaft:** The rotor turns the low-speed shaft at about 15 to 30 rotations per minute.
- **Gear box** connect the low-speed shaft to the high-speed shaft and increases the rotational speeds from about **15 to 30 rotations per minute (rpm)** to about **1000 to 1800 rpm**, the rotational speed required by most generators (alternators) to produce electricity. This is an expensive and heavy part of wind turbines.
- **Generator** is usually an induction generator that produces **50-cycle AC electricity**.
- **Controller** starts up the machine at wind speeds of about **8 to 16 miles per hour (mph)** and shuts off the machine at about **55 mph**. Turbines do not operate at wind speeds above about **55 mph** because that might damage them.
- **Anemometer:** It measures the wind speed and transmits the data to the controller. The controller then corrects the turbine's direction, pitch, and yaw to best harvest the available wind energy.